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Authors:	G. White	T. Fossati	R. Geib
	<i>CableLabs</i>	<i>Linaro</i>	<i>Deutsche Telekom</i>

RFC 9956

A Non-Queue-Building Per-Hop Behavior (NQB PHB) for Differentiated Services

Abstract

This document specifies characteristics of a Non-Queue-Building Per-Hop Behavior (NQB PHB). The NQB PHB provides a shallow-buffered, best-effort service as a complement to a Default deep-buffered, best-effort service for Internet services. The purpose of this NQB PHB is to provide a separate queue that enables smooth (i.e., non-bursty), low-data-rate, application-limited traffic microflows, to avoid the delay, delay variation and loss that would ordinarily be caused by sharing a queue with bursty, capacity-seeking traffic. This PHB is implemented without prioritization and can be implemented without rate policing, making it suitable for environments where the use of these features is restricted. The NQB PHB has been developed primarily for use by access network segments, where queuing delay and queuing loss caused by Queue-Building (QB) protocols are manifested; however, its use is not limited to such segments. In particular, the application of NQB PHB to cable broadband links, Wi-Fi links, and mobile network radio/core segments are discussed in this document. This document recommends a specific Differentiated Services Code Point (DSCP) to identify NQB microflows and updates the guidance in RFC 8325 on mapping Differentiated Services (Diffserv or DS) to IEEE 802.11 for this codepoint.

Status of This Memo

This is an Internet Standards Track document.

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1. Introduction

This document defines a Diffserv PHB called the "Non-Queue-Building Per-Hop Behavior" (or "NQB PHB"). The NQB PHB isolates traffic microflows (application-to-application flows, see [Section 1.2](#) of [\[RFC2475\]](#)) that have relatively low data rates and that do not, themselves, materially contribute to queuing delay and loss. This isolation allows these traffic microflows to avoid the queuing delay and losses caused by other traffic.

NQB microflows such as interactive voice, game sync packets, certain machine-to-machine applications, DNS lookups, and some real-time Internet of Things (IoT) analytics data are low-data-rate, application-limited microflows. These can be distinguished from bursty traffic microflows and high-data-rate traffic microflows managed by a classic congestion control algorithm (both of which cause queuing delay and loss). The term "classic congestion control" is defined in [\[RFC9330\]](#) to mean congestion control that coexists with standard Reno congestion

control [RFC5681]. In this document, we will use "Queue-Building" (or "QB") to refer to microflows that cause queuing delay and loss. See [Section 1.2](#) of [RFC2475] for definitions of other terminology used in this document.

In accordance with IETF guidance in [RFC2914] and [RFC8085], most packets carried by access networks are managed by an end-to-end congestion control algorithm. Many of the commonly deployed congestion control algorithms, such as Reno [RFC5681], CUBIC [RFC9438], or BBR [BBR-CC], are designed to seek the available capacity of the path from sender to receiver (which can frequently be the access network link capacity). In doing so, they generally overshoot the available capacity, causing a queue to build up at the bottleneck link. This queue buildup results in variable delay and packet loss that can affect all the applications that are sharing the bottleneck link. Moreover, many bottleneck links implement a relatively deep buffer (100 ms or more) (see [Gettys2012], [Cardwell2017], [WikipediaBufferbloat], and [BBR-CC]) in order to enable these congestion control algorithms to use the link efficiently, which exacerbates the delay and delay variation experienced.

In contrast to applications that frequently cause queuing delay, there are a variety of relatively low data rate applications that do not materially contribute to queuing delay and loss; nonetheless, they are subjected to it by sharing the same bottleneck link. Many of these applications can be sensitive to delay or delay variation, as well as packet loss; thus, they produce a poor Quality of Experience (QoE) in such conditions.

Active Queue Management (AQM) mechanisms intended for single queues (such as Proportional Integral Controller Enhanced (PIE) [RFC8033], DOCSIS-PIE [RFC8034], PI2 [RFC9332], or CoDel [RFC8289]) can improve the QoE for delay-sensitive applications, but there are practical limits to the amount of improvement that can be achieved without impacting the throughput of capacity-seeking applications. For example, AQMs generally allow a significant amount of queue depth variation to accommodate the behaviors of congestion control algorithms such as Reno and CUBIC. If the AQM attempted to control the queue depth much more tightly, applications using those algorithms would not fully utilize the link. Alternatively, flow-queuing systems, such as fq_codel [RFC8290], can be employed to isolate microflows from one another; however, they are not appropriate for all bottleneck links due to reasons that include complexity.

The NQB PHB supports differentiating between these two classes of traffic in bottleneck links and queuing them separately so that both classes can deliver satisfactory QoE for their applications. In particular, the NQB PHB provides a shallow-buffered, best-effort service as a complement to a Default (see [RFC2474]) deep-buffered, best-effort service. This PHB is designed for broadband access network links (where there is minimal aggregation of traffic), especially when buffers are deep (see [Section 3.4](#)). The applicability of this PHB to lower-rate links is discussed in [Section 6.6](#).

To be clear, a network implementing the NQB PHB solely provides isolation for traffic classified as behaving in conformance with the behaviors discussed in [Section 3.1](#). A node supporting the NQB PHB makes no guarantees on delay or data rate for NQB-marked microflows (beyond the delay bound provided by the shallow buffer), it is the NQB senders' behavior itself that results in low delay and low loss.

Sections 6 and 7 of this document provide guidance for network operators regarding appropriate forwarding of traffic marked with the NQB DSCP. The guidance includes recommendations for the configuration of network nodes that support the NQB PHB as well as for network nodes that do not support the PHB; this allows NQB traffic to be forwarded in a way that aligns with the goals for NQB treatment and supports the use of this codepoint by other nodes and other networks.

2. Requirements Language

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

3. Context

3.1. NQB Behavior

There are applications that send traffic at relatively low data rates and/or in a fairly smooth and consistent manner such that they are unlikely to exceed the available capacity of the network path between sender and receiver, even at an inter-packet timescale. Some of these applications are transactional in nature; they might only send one packet (or a few packets) per RTT. These applications might themselves only cause very small, transient queues to form in network buffers; nonetheless, they can be subjected to delay and delay variation as a result of sharing a network buffer with applications that tend to cause large and/or standing queues to form. These applications typically implement a response to network congestion that consists of discontinuing (or significantly reducing) transmissions. These applications can be negatively affected by excessive delay and delay variation. Such applications are ideal candidates to be queued separately from the applications that are the cause of queue buildup, delay, and loss.

In contrast, QB microflows include those that probe for link capacity and induce delay and loss as a result, for example, microflows that use CUBIC, Reno, or other TCP/QUIC congestion control algorithms in a capacity-seeking manner. Other types of QB microflows include those that send at a high burst rate even if the long-term average data rate is much lower.

3.2. Relationship to the Diffserv Architecture

The IETF has defined the Diffserv architecture [RFC2475] with the intention that it allows traffic to be marked in a manner that conveys the performance requirements of that traffic either qualitatively or in a relative sense (e.g., priority). The architecture defines the use of the DSCP field [RFC2474] for this purpose, and numerous RFCs have been written that describe recommended interpretations of the values (Diffserv Codepoints [RFC2474]) of the field, and standardized treatments (traffic conditioning and per-hop-behaviors [RFC2475]) that can be implemented to satisfy the performance requirements of traffic so marked.

While this architecture is powerful and flexible enough to be configured to meet the performance requirements of a variety of applications and traffic categories, or to achieve differentiated service offerings, meeting the performance requirements of an application across the entire sender-to-receiver path involves all the networks in the path agreeing on what those requirements are and sharing an interest in meeting them. In many cases, this is made more difficult since the performance "requirements" are not strict ones (e.g., applications will degrade in some manner as loss, delay, and/or delay-variation increase), so the importance of meeting them for any particular application in some cases involves a judgment as to the value of avoiding some amount of degradation in quality for that application in exchange for an increase in the degradation of another application.

Further, in many cases, the implementation of Diffserv PHBs has historically involved prioritization of service classes with respect to one another, which sets up the zero-sum game alluded to in the previous paragraph and which results in the need to limit access to higher priority classes via mechanisms such as access control, admission control, traffic conditioning and rate policing, and/or to meter and bill for carriage of such traffic. These mechanisms can be difficult or impossible to implement in the Internet.

In contrast, the NQB PHB has been designed with the goal that it avoids many of these issues; thus, it could conceivably be deployed across the Internet. The intent of the NQB DSCP is that it signals verifiable behavior that permits the sender to request differentiated treatment. Also, the NQB traffic is to be given a separate queue with forwarding preference equal to Default traffic and given no reserved capacity other than any minimum capacity that it shares with Default traffic. As a result, the NQB PHB does not aim to meet specific application performance requirements. Instead, the sole goal of the NQB PHB is to isolate NQB traffic from other traffic that causes an increase in loss, delay, and/or delay-variation, given that the NQB traffic is, itself, only an insignificant contributor to those degradations. The PHB is also designed to reduce the incentives for a sender to mis-mark its traffic since neither higher capacity nor reserved capacity are being offered. These attributes eliminate many of the trade-offs that underlie the handling of differentiated service classes in the Diffserv architecture as it has previously been defined. These attributes also significantly simplify access control and admission control functions, reducing them to simple verification of behavior. This aspect is discussed further in Sections 4 and 5.2.

Therefore, the NQB PHB is intended for the situation where the performance requirements of applications cannot be assured across the whole sender-to-receiver path; as a result, applications cannot feasibly place requirements on the network. Instead, many applications have evolved to make the best out of the network environment that they find themselves in. In this context, the NQB PHB is intended to provide a better network environment for applications that send data at relatively low and non-bursty data rates.

In regard to a comparison between the NQB PHB and other standardized PHBs in the Diffserv series, the closest similarity is to the Expedited Forwarding (EF) PHB [RFC3246], which also intends to enable services that provide low loss, low delay, and low-delay variation. Unlike EF, NQB has no requirement for a guaranteed minimum rate, nor does it have a requirement to police incoming traffic to such a rate: NQB is expected to be given the same forwarding preference as Default traffic. See [Appendix B](#) for a more detailed comparison of the NQB and EF PHBs.

In nodes that support multiple Diffserv Service Classes, NQB traffic is intended to be handled as a part of the Default treatment. Traffic assigned to this class does not receive better forwarding treatment (e.g., prioritization) with respect to other classes, AFxx, EF, etc. Of course, traffic marked as NQB could (like other Default traffic) receive better forwarding treatment with respect to Lower-Effort (LE) [RFC8622] (e.g., the NQB queue could be emptied in a priority sequence before the LE queue).

3.3. Relationship to L4S

In this document, the NQB DSCP and PHB have been defined to operate independently of the Low Latency, Low Loss, and Scalable throughput (L4S) architecture [RFC9330]. Nonetheless, traffic marked with the NQB DSCP is intended to be compatible with L4S [RFC9330], with the result being that NQB traffic and L4S traffic can share the low-latency queue in an L4S DualQ node [RFC9332]. A network node's compliance with the DualQ Coupled AQM requirements (see Section 2.5 of [RFC9332]) is considered sufficient to support the NQB PHB requirement of fair allocation of capacity between the QB and NQB queues (Section 5). Note that these requirements, in turn, require compliance with all the requirements in Section 5 of [RFC9331].

Applications that comply with both the NQB sender requirements in Section 4 and the "Prague L4S requirements" in Section 4 of [RFC9331] could mark their packets both with the NQB DSCP and with the ECT(1) value.

In nodes that support both the NQB PHB and L4S, the L4S network functions **SHOULD** treat packets marked with the NQB DSCP and ECT(1) or Congestion Experienced (CE) the same as packets marked with the Default DSCP and the same Explicit Congestion Notification (ECN) value. Here, "L4S network functions" refers to the L4S Network Node functions (see Section 5 of [RFC9331]), and any mechanisms designed to protect the L4S queue (such as those discussed in Section 8.2 of [RFC9330]). The processing by an L4S node of an ECT(0) packet that is classified to the L queue (e.g., as a result of being marked with a NQB DSCP) is specified in Section 5.4.1.1 of [RFC9331] and Section 2.5.1.1 of [RFC9332].

Additionally, Section 5.4 places requirements on treatment of ECN-marked packets by a node that supports the NQB PHB.

3.4. Applicability

This PHB is primarily applicable for high-speed broadband access network links, where there is minimal aggregation of traffic and deep buffers are common.

In many other links, forwarding NQB-marked packets using the Default treatment might be sufficient to preserve the loss, delay, and delay-variation performance for NQB traffic. This is generally true in links that do not typically experience congestion (for example, many backbone and core network links) and in highly aggregated links (links designed to carry a large number of simultaneous microflows) where individual microflow burstiness is averaged out and, thus, is unlikely to cause much actual delay. Section 6.2 provides recommendations for configuration of network nodes in such cases.

4. NQB Sender Requirements

Microflows that are eligible to be marked with the NQB DSCP are ones that send non-bursty traffic at a low data rate relative to typical network path capacities. Here, the data rate is limited by the application itself rather than by network capacity: these microflows send at a data rate of no more than about 1% of the "typical" network path capacity. In addition, these microflows are required to be sent in a smooth (i.e., paced) manner, where the number of IP bytes sent in any time interval "T" is less than or equal to $(R * T) + MTU$, where "R" is the maximum rate described in the preceding sentence, expressed in bytes-per-second. For example, at the time of writing, access network data rates are typically on the order of 50 Mbps or more in the Internet (see [Section 6.6](#) for a discussion of cases where this isn't true): this implies 500 kbps as an upper limit. Note that microflows are unidirectional, while application traffic is often bidirectional (i.e., an application instance might consist of one or more microflows in each direction). For a particular application, it could be the case that some of its microflows are eligible to be marked with the NQB DSCP while others are not. For example, an interactive video streaming application might consist of a high-bandwidth video stream (not eligible for NQB marking) in one direction and a low-bandwidth control stream (eligible for NQB marking) in the other.

Microflows marked with the NQB DSCP are expected to comply with existing guidance for safe deployment on the Internet, including the guidance related to response to network congestion, for example the requirements in [\[RFC8085\]](#) and [Section 2](#) of [\[RFC3551\]](#) (also see the circuit breaker limits in [Section 4.3](#) of [\[RFC8083\]](#) and the description of inelastic pseudowires in [Section 4](#) of [\[RFC7893\]](#)). The fact that a microflow's data rate is low relative to typical network capacities is no guarantee that sufficient capacity exists in any particular network, and it is the responsibility of the application to detect and react appropriately if the network capacity is insufficient. To be clear, the description of NQB-marked microflows in this document is not to be interpreted as suggesting that applications generating such microflows are in any way exempt from this responsibility. One way that an application marking its traffic as NQB can handle this is to implement a scalable congestion control mechanism as described in [\[RFC9331\]](#).

The DS field specification requires the definition of a recommended DSCP to be associated with each standardized PHB (see [Section 5](#) of [\[RFC2474\]](#)). In accordance with this, applications are **RECOMMENDED** to use the DSCP value 45 (decimal) to mark microflows as NQB. The choice of the DSCP value 45 (decimal) is motivated, in part, by the desire to achieve separate queuing in existing Wi-Fi networks (see [Section 7.3](#)) and by the desire to make implementation of the PHB simpler in network equipment that has the ability to classify traffic based on ranges of DSCP values (see [Section 6.3](#) for further discussion).

The two primary considerations for whether an application chooses to mark its traffic as NQB involve the risks of being subjected to a traffic protection algorithm (see [Section 5.2](#)) and/or to the consequences of overrunning the NQB shallow buffer if (in either case) the traffic contributes to the formation of a queue in a node that supports the PHB. In both cases, the result could be that excess traffic is discarded or queued separately as Default traffic (and, thus, potentially is delivered out of order). To avoid these risks, if a microflow's traffic exceeds the rate equation

provided in the first paragraph of this section, the application **MUST NOT** mark this traffic with the NQB DSCP. In such a case, the application could instead consider using a scalable congestion control mechanism as described in [\[RFC9331\]](#).

The sender requirements outlined in this section are all related to observable attributes of the packet stream, which makes it possible for network elements (including nodes implementing the PHB) to monitor for inappropriate usage of the DSCP and take action (such as discarding or re-marking) on traffic that does not comply. This functionality, when implemented as part of the PHB, is described in [Section 5.2](#).

5. NQB PHB Requirements

For the NQB PHB to become widely deployed, it is important that incentives are aligned correctly, i.e., that there is a benefit to the application in marking its packets correctly and a disadvantage (or at least no benefit) to an application in intentionally mis-marking its traffic. Thus, a useful property of nodes (i.e., network switches and routers) that support separate queues for NQB and QB microflows is that each queue tends to be the better choice for the traffic it is designed for: the NQB queue for microflows consistent with the NQB sender requirements in [Section 4](#) and the QB queue for those that are not. By adhering to these principles, there is little incentive for senders to mis-mark their traffic as NQB.

This principle of incentive alignment ensures a system is robust to the behavior of the large majority of individuals and organizations who can be expected to act in their own interests (including application developers and service providers who act in the interests of their users). Malicious behavior is not necessarily based on rational self-interest, so incentive alignment is not a sufficient defense, but the large majority of users do not act out of malice. Protection against malicious attacks (and accidents) is addressed in [Section 5.2](#) and summarized in [Section 9](#). As mentioned previously, the NQB designation and marking is intended to convey verifiable traffic behavior, as opposed to simply a desire for differentiated treatment. As a result, any mis-marking can be identified by the network.

5.1. Primary Requirements

A node supporting the NQB PHB **MUST** provide a queue for NQB traffic separate from the queue used for Default traffic.

A node supporting the NQB PHB **SHOULD NOT** rate limit or rate police the aggregate of NQB traffic separately from Default traffic. An exception to this recommendation for traffic sent towards a non-DS-capable domain is discussed in [Section 6.4.1](#). Note also that [Section 5.2](#) discusses potential uses of per-microflow (rather than aggregate) rate policing.

The NQB queue **SHOULD** be given equivalent forwarding preference compared to the Default queue. The node **SHOULD** provide a scheduler that allows NQB and Default traffic to share the link in a manner that treats the two classes equally, e.g., a Deficit Round-Robin (DRR) scheduler with equal weights or two Wireless Multimedia Access Categories with the same Enhanced Distributed Channel Access (EDCA) parameters. The use of equal weights for DRR is given as a reasonable example and is not intended to preclude other scheduling weights (see below for

details). A node that provides rate limits or rate guarantees for Default traffic **SHOULD** ensure that such limits and/or guarantees are shared with NQB traffic in a manner that treats the two classes equally. This could be supported using a hierarchical scheduler where the rate limits and guarantees are configured on a parent class, and the two queues (Default and NQB) are arranged as the children of the parent class and given equal access to the capacity configured for the parent class (e.g., with equal DRR scheduling). Compliance with these recommendations reduces the incentives for QB traffic to be mis-marked as NQB and is most important in nodes that are likely bottlenecks, where deviation from them could result in a discernible benefit for mis-marked traffic (to the detriment of other traffic). In network nodes that are rarely bottlenecks, these recommendations are less critical.

In the DRR example above, equal scheduling weights is only an example. Ideally, the DRR weight would be chosen to match the highest fraction of capacity that NQB-compliant flows are likely to use on a particular network segment. Given that NQB-compliant flows are not capacity seeking (in contrast to QB flows, which can be), and since DRR allows unused capacity in one class to be used by traffic in the other, providing a higher-than-needed NQB scheduler weight could be considered less problematic than the reverse. That said, providing a higher-than-needed NQB scheduler weight does increase the likelihood that a non-compliant microflow mis-marked as NQB is able to use more than its fair share of network capacity. NQB microflows are expected to each consume no more than 1% of the link capacity, and in low stat-mux environments (such as at the edge of the network) would be unlikely in aggregate to consume 50% of the link capacity. Thus, 50% seems a reasonable upper bound on the weight for the NQB PHB in these environments.

By default, a node supporting the NQB PHB **SHOULD** classify packets marked with the DSCP value 45 (decimal) into the queue for NQB traffic. In accordance with the requirement in [Section 3 of \[RFC2474\]](#), a node supporting the NQB PHB **MUST** support the ability to configure the DSCP that is used to classify packets into the queue for NQB traffic. A node supporting the NQB PHB **MAY** support the ability to configure multiple DSCPs that are used to classify packets into the queue for NQB traffic.

Support for the NQB PHB is advantageous at bottleneck nodes. Many bottleneck nodes have a relatively deep buffer for Default traffic (e.g., roughly equal to the base RTT of the expected connections, which could be tens or hundreds of milliseconds). Providing a similarly deep buffer for the NQB queue would be at cross purposes to providing very low queueing delay and would erode the incentives for QB traffic to be marked correctly at such a bottleneck node. The NQB queue **MUST** have a buffer size that is significantly smaller than the buffer provided for Default traffic. It is **RECOMMENDED** to configure an NQB buffer size less than or equal to 10 ms at the shared NQB/Default egress rate.

In order to enable network operators to monitor the usage of the NQB PHB and, in particular, to monitor for potential mis-marking of QB traffic, a node supporting the NQB PHB **MUST** provide statistics that can be used by the network operator to detect whether abuse is occurring (e.g., packet and drop counters). Support for such counters ensures that operators who configure the NQB PHB have the ability to track the amount of packet drop that is occurring due to traffic overrunning the shallow buffer and then take action if they believe the PHB is causing more issues than it is solving in their environment. Those actions could include disabling the PHB,

identifying and dealing with the sources of malicious traffic directly, enabling traffic protection (Section 5.2) if it is available, or pursuing a feature request with the equipment manufacturer to add a traffic protection function if it isn't currently available.

To prevent propagation of degradation of service for NQB traffic caused by potential mis-marking of QB traffic, network equipment that supports this PHB and handles traffic for multiple users **SHOULD** support provisioning of capacity and related forwarding resources on a per-user basis and **SHOULD** support enforcement of the resulting per-user limits on the aggregate of NQB and QB traffic for each user. In this context, the term "user" should be read as meaning an entity that the equipment is designed to serve more-or-less independently, such as a subscriber in the case of access network equipment, a station (STA) in the case of a Wi-Fi Access Point (AP) that implements per-STA queuing and airtime fairness, or an end user in the case of a networking co-op that serves a set of end users. This functionality is commonly available in the class of network equipment for which this PHB is primarily applicable (see Section 3.4). Provisioning methodology as well as decisions on whether and how to enforce the resulting limits may vary by network operator.

While not fully described in this document, it may be possible for network equipment to implement a separate QB/NQB pair of queues for additional service classes beyond the Default PHB / NQB PHB pair.

In some cases, existing network equipment has been deployed that cannot readily be upgraded or configured to support the PHB requirements. However, this equipment might be capable of loosely supporting an NQB service; see Section 7.3.1 for details and an example where this is particularly important. A similar approach might prove to be useful in other network environments.

5.2. Traffic Protection

It is possible that, due to an implementation error or misconfiguration, a QB microflow could end up being mis-marked as NQB or vice versa. It is also possible that a malicious actor could introduce a QB microflow marked as NQB with the intention of causing disruptions. In the case of a low-data-rate microflow that isn't marked as NQB and therefore ends up in the QB queue, it would only impact its own Quality of Service (QoS); therefore, it seems to be of lesser concern. However, a QB microflow that is mis-marked as NQB is able to contribute to NQB queue formation at a network node that would cause queuing delay and/or loss for all the other microflows that are sharing the NQB queue.

To prevent this situation from harming the performance of the microflows that comply with the requirements in Section 4, network elements that support the NQB PHB **SHOULD** support a "traffic protection" function that can identify microflows or packets that are inconsistent with the sender requirements in Section 4 and either reclassify those microflows/packets to the QB queue or discard the offending traffic. In the case of a traffic protection algorithm that reclassifies offending traffic, the implementation **MAY** provide an option to additionally re-mark such traffic to Default (or possibly to another local use codepoint) so that the result of the traffic protection

decision can be used by further hops. This sort of re-marking could provide a limited layer of protection in situations where downstream network nodes support separate queuing for NQB-marked packets but lack support for traffic protection.

If traffic protection is not supported or is not effective in preventing queue formation and growth in the NQB queue, then QB traffic that is mis-marked as NQB is able to form a queue that overflows the shallow buffer provided for NQB traffic; this is expected to result in redirecting the excess packets to the QB queue or discarding them. Both actions degrade service for not only the mis-marked QB traffic, but also for any correctly marked NQB traffic. This will likely cause a significant degradation of service for NQB traffic. Even if mis-marked QB traffic does not cause buffer overflow, the queue that forms results in QB traffic obtaining the reduced loss and delay benefits of the NQB service while causing queuing delay for all the other microflows that are sharing the queue. These increased abilities of QB traffic to damage the NQB service in the absence of a traffic protection function need to be considered. This is the motivation for the "**SHOULD**" requirement to support traffic protection (in the previous paragraph). An NQB PHB implementation that does not support traffic protection risks being limited to deployment situations where traffic protection is potentially not necessary. One example of such a situation could be a controlled environment (e.g., enterprise LAN) where a network administrator is expected to manage the usage of DSCPs.

As it is defined here, traffic protection differs from Traffic Conditioning implemented in other Diffserv contexts. Traffic Conditioning is commonly performed at the edge of a DS domain (either ingress or egress, depending on Traffic Conditioning Agreements (TCAs) in place). In contrast, traffic protection is intended to be implemented in the nodes that implement the PHB. By placing the traffic protection at the PHB node, an implementation can monitor the actual NQB queue and take action only if a queue begins to form. Implementation of traffic protection at PHB nodes that are most likely to be a bottleneck is particularly important because these are the nodes that would be expected to show the most queue buildup in the presence of QB traffic mis-marked as NQB.

This specification does not mandate a particular algorithm for traffic protection. This is intentional since this will probably be an area where implementers innovate. The specifics of traffic protection could need to be different in different network equipment and in different network contexts. Instead, this specification provides guidelines and some examples of traffic protection algorithms that could be employed.

The traffic protection function **SHOULD NOT** base its decisions upon application-layer constructs (such as the port number used by the application or the source/destination IP address). Instead, it ought to base its decisions on the actual behavior of each microflow (i.e., the pattern of packet arrivals).

A conventional implementation of such a traffic protection algorithm is a per-microflow rate policer, designed to identify microflows that exceed the bound provided in [Section 4](#), where the value R is set to 1% of the egress link capacity available for NQB traffic. An alternative is to use a traffic protection algorithm that bases its decisions on the detection of actual queuing (i.e., by monitoring the queuing delay experienced by packets in the NQB queue) in correlation with the arrival of packets for each microflow. While a per-microflow rate policer is conceptually simpler

(and is based directly on the NQB sender requirements), it could often end up being more strict than is necessary (for example, by policing a flow that exceeds the rate equation even when the link is underutilized). One example traffic protection algorithm based on the detection of actual queuing can be found in [\[RFC9957\]](#). This algorithm maintains per-microflow state for a certain number of simultaneous QB microflows (e.g., 32), and shared state for any additional microflows above that number.

In the case of a traffic protection algorithm that reclassifies offending traffic, different levels of hysteresis could be considered. For example, the reclassify decision could be made on a packet-by-packet basis, which could result in significant out-of-order delivery for offending microflows as some portion of the microflow's packets remain in the NQB queue and some are reclassified to the Default queue. Alternatively, a traffic protection function could employ a certain level of hysteresis to prevent borderline microflows from being reclassified capriciously, thus causing less potential for out-of-order delivery. As a third option, the decision could be made to take action on all the future packets of the microflow, though sufficient logic would be needed to ensure that a future microflow (e.g., with the same 5-tuple) isn't misidentified as the current offending microflow.

In the case of a traffic protection algorithm that discards offending traffic, similar levels of hysteresis could be considered. In this case, it is **RECOMMENDED** that the decision thresholds be set higher than in the case of designs that reclassify since the degradation of communications caused by the packet being discarded is likely to be greater than the degradation caused by out-of-order delivery.

The traffic protection function described here might require that the network element maintain microflow state. The traffic protection function **MUST** be designed such that the node implementing the NQB PHB does not fail (e.g., crash) in the case that the microflow state is exhausted. This might be accomplished simply by controlling/limiting the resources dedicated to tracking misbehaving flows.

Some networks might prefer to implement a Traffic Conditioning approach that polices the application of the NQB DSCP at the ingress edge so that per-hop traffic protection is not needed. This could be accomplished via the use of a per-microflow rate policer that polices microflows at 1% of the minimum link capacity of the network. This approach would generally be expected to be inferior to per-hop traffic protection because:

- on one hand, it would be difficult for edge nodes to guarantee that there would never be more than 100 NQB flows that would share a single internal bottleneck, and
- on the other hand, there could be internal links that have much greater capacity than the minimum.

So, Traffic Conditioning at the edge could simultaneously be too lenient and too strict.

5.3. Limiting Packet Bursts from Links

Some link technologies introduce burstiness by briefly storing packets prior to forwarding them. A common cause of this burstiness is link discontinuity (i.e., where the link is not continuously available for transmission by the device), for example, time-division-duplex links or time-division-multiple-access (TDMA) links. Some link technologies that fall into this category are Passive Optical Networks (PONs), Wi-Fi, LTE/5G, and Data-Over-Cable Service Interface Specification (DOCSIS).

As well as NQB senders needing to limit packet bursts (see [Section 4](#)), traffic designated for the NQB PHB would benefit from configuring these link technologies to limit the burstiness introduced. This is for three reasons:

1. Burstiness, whether caused by the sender or by a link on the path, could cause queuing delay at downstream bottlenecks and, thus, degrade QoE.
2. Burstiness in links typically means that packets have been delayed by a variable amount. That is, for packets that are being aggregated awaiting a transmission opportunity, some packets would generally have arrived just after the last transmission opportunity and would have to wait the longest while others would generally arrive just in time for the next transmission opportunity and would wait the least. This manifests as delay variation that can also degrade QoE for applications that desire NQB treatment.
3. A downstream bottleneck that implements the NQB PHB could have implemented a traffic protection mechanism ([Section 5.2](#)) that responds to queuing delay by re-marking/reclassifying/dropping packets. Bursty arrivals caused by an upstream link could introduce queuing delay in the NQB queue and these would be more likely to be subjected to traffic protection effects.

This document does not set any quantified requirements for links to limit bursts; this is primarily because link technologies are outside the remit of Diffserv specifications. However, it would not seem necessary to limit bursts lower than roughly 10% of the minimum base RTT expected in the typical deployment scenario (e.g., 250 μ s burst duration for links within the public Internet). This observation aligns with a similar one in [Section 5.5](#) of [\[RFC9331\]](#).

5.4. Treatment of the Explicit Congestion Notification Field

NQB network functions **MUST** treat packets marked with the NQB DSCP uniformly, regardless of the value of the ECN field. Here, the term "NQB network functions" refers to the traffic protection function (defined in [Section 5.2](#)) and any re-marking/traffic policing function designed to protect unmanaged networks (as described in [Section 6.4.1](#)).

6. Operational Considerations

This section describes considerations for network operators regarding the identification, marking, and handling of NQB traffic. It outlines aggregation behaviors and special considerations in unmanaged and inter-network scenarios. Additionally, [Section 6.2](#) contains

configuration recommendations for nodes that do not support the NQB PHB. [Section 6.4.1](#) contains configuration recommendations for networks that interconnect with non-DS-capable domains.

6.1. NQB Traffic Identification

As required in [Section 5](#), nodes supporting the NQB PHB provide for the configuration of classifiers that can be used to differentiate between QB and NQB traffic of equivalent importance. The assigned NQB DSCP value (45 decimal) is recommended for use as the default classifier to distinguish NQB traffic from traffic classified as Default (DSCP 0) (see [Sections 4](#) and [5.1](#)).

6.2. Aggregation of the NQB DSCP into Another Diffserv PHB

It is **RECOMMENDED** that networks and nodes that do not support the NQB PHB be configured to treat traffic marked with the NQB DSCP the same as traffic with the Default DSCP. This includes networks (such as MPLS) and nodes that aggregate service classes as discussed in [\[RFC5127\]](#) and [\[RFC8100\]](#); in this case, this recommendation would result in traffic marked with the NQB DSCP being aggregated into the Elastic Treatment Aggregate (for networks as described in [\[RFC5127\]](#)) or the Default / Elastic Treatment Aggregate (for networks as described in [\[RFC8100\]](#)).

Networks and nodes that do not support the NQB PHB ought to only classify packets with the NQB DSCP value into the appropriate treatment aggregate, or encapsulate such packets for purposes of aggregation, and **SHOULD NOT** re-mark them with a different DSCP. This preservation of the NQB DSCP value enables hops further along the path to provide the NQB PHB successfully. This aligns with recommendations in [\[RFC5127\]](#).

In nodes that do not typically experience congestion (for example, many backbone and core network switches), forwarding packets with the NQB DSCP using the Default treatment might be sufficient to preserve the loss, delay, and delay-variation performance for NQB traffic.

In nodes that do experience congestion, forwarding packets with the NQB DSCP using the Default treatment could result in degradation of the loss, delay, and delay-variation performance but nonetheless preserves the incentives described in [Section 5](#).

Aggregating traffic marked with the NQB DSCP into a PHB designed for real-time, delay-sensitive traffic (e.g., the Real-Time Treatment Aggregate [\[RFC5127\]](#) or the Bulk Real-Time Treatment Aggregate [\[RFC8100\]](#)), might better preserve the loss, delay, and delay-variation performance in the presence of congestion; however, it would need to be done with consideration of the risk of creating an incentive for non-compliant traffic to be mis-marked as NQB.

6.3. Aggregation of Other DSCPs into the NQB PHB

The Diffserv model provides flexibility for operators to control the way they choose to aggregate traffic marked with a specific DSCP. Operators of nodes that support the NQB PHB could choose to aggregate other service classes into the NQB queue. This is particularly useful in cases where specialized PHBs for these other service classes had not been provided at a potential bottleneck, perhaps because it was too complex to manage traffic contracts and conditioning. Candidate service classes for this aggregation would include those that carry low-data-rate inelastic traffic

that has low to very-low tolerance for loss, delay, and/or delay variation. Operators would need to use their own judgment based on the actual traffic characteristics in their networks in deciding whether or not to aggregate other service classes / DSCPs with NQB. For networks that use the service class definitions from [\[RFC4594\]](#), this could include Telephony (EF/VA), Signaling (CS5), and possibly Real-Time Interactive (CS4) (depending on data rate). In the preceding sentence, "VA" and "CSx" refer to VOICE-ADMIT ([\[RFC5865\]](#)) and Class Selector ([\[RFC2474\]](#)), respectively. In some networks, equipment limitations may necessitate aggregating a range of DSCPs (e.g., traffic marked with DSCPs 40-47 (decimal), i.e., those whose three most significant bits are 0b101). As noted in [Section 4](#), the choice of the DSCP value 45 (decimal) is motivated in part by the desire to make this aggregation simpler in network equipment that can classify packets via comparing the DSCP value to a range of configured values.

A node providing only a NQB queue and a Default queue may obtain an NQB performance similar to that of EF, for example, as described by [Appendix A.3.1](#) of [\[RFC2598\]](#). Some caveats and differences are discussed in [Appendix B](#).

6.4. Cross-Domain Usage and DSCP Re-marking

In contrast to some existing standard PHBs, which are typically only used within a DS domain (e.g., an Autonomous System (AS) or an enterprise network), this PHB is expected to be used across the Internet, wherever suitable operator agreements apply. Under the model described in [\[RFC2474\]](#), this requires that the corresponding DSCP is recognized and mapped across network boundaries accordingly.

If NQB support is extended across a DS domain boundary, the interconnected networks agreeing to support NQB **SHOULD** use the DSCP value 45 (decimal) for NQB at network interconnection, unless a different DSCP is explicitly documented in the TCA (see [\[RFC2475\]](#)) for that interconnection. If a Diffserv-Intercon Interconnection Class (see [Section 4](#) of [\[RFC8100\]](#)) is operational between interconnected domains, the receiving domain may prefer a different DSCP for NQB traffic that allows for a DSCP range-based classification for the Default / Elastic Treatment Aggregate. Similar to the handling of DSCPs for other PHBs (and as discussed in [\[RFC2475\]](#)), networks can re-mark NQB traffic to a DSCP value other than 45 (decimal) for internal usage. To ensure reliable NQB PHB treatment on the entire path, the appropriate NQB DSCP would need to be restored when forwarding to another network.

6.4.1. Interoperability with Non-DS-Capable Domains

As discussed in [Section 4](#) of [\[RFC2475\]](#), there may be cases where a network operator that supports Diffserv is delivering traffic to another network domain (e.g., a network outside of their administrative control) where there is an understanding that the downstream domain does not support Diffserv or there is no knowledge of the traffic management capabilities of the downstream domain, and no agreement in place. In such cases, [Section 4](#) of [\[RFC2475\]](#) suggests that the upstream domain opportunistically re-mark traffic with a Class Selector Codepoint or DSCP 0 (Default) under the assumption that traffic so marked would be handled in a predictable way by the downstream domain.

In the case of a network that supports the NQB PHB (and carries traffic marked with the recommended DSCP value 45 (decimal)), the same concerns apply. In particular, since the recommended NQB DSCP value 45 (decimal) could be given high priority in some non-DS-compliant network equipment (e.g., legacy Wi-Fi APs as described in [Section 7.3.1](#)), it is **RECOMMENDED** that the operator of the upstream domain implement one of the following safeguards before delivering traffic into a non-DS-capable domain:

1. One option for such a safeguard is to re-mark NQB traffic to DSCP 0 (Default) (or another Class Selector DSCP) before delivering traffic into a non-DS-capable domain, in accordance with the suggestion in [Section 4](#) of [RFC2475]. Network equipment designed for such environments **SHOULD**, by default, re-mark NQB traffic to DSCP 0 (Default) and **SHOULD** support the ability to change and disable this re-marking. Re-marking NQB traffic to DSCP 0 (Default) could be considered the "safest" approach since the upstream domain can thereby ensure that NQB traffic is not given inappropriate treatment in the non-DS-capable domain. That said, it comes with the downside that the re-marking ruins any possibility of NQB isolation in any further downstream domain (not just the immediate neighbor).
2. As an alternative to re-marking all NQB traffic, such an operator could deploy a traffic protection (see [Section 5.2](#)) or a shaping/policing function on traffic marked with the NQB DSCP that minimizes the potential for negative impacts on Default traffic, should the downstream domain treat traffic with the NQB DSCP as high priority.

In the case that a traffic protection function is used, it **MUST** either re-mark offending traffic to DSCP 0 (or another Class Selector DSCP) or discard it. Note that a traffic protection function, as defined in this document, might only provide protection from issues occurring in subsequent network hops if the device implementing the traffic protection function is the bottleneck link on the path, so it might not be a solution for all situations.

In the case that a traffic policing function or a rate-shaping function is applied to the aggregate of NQB traffic destined to such a downstream domain, the policer/shaper rate **SHOULD** be set to either 5% of the interconnection data rate or 5% of the typical rate for such interconnections, whichever is greater, with excess traffic being re-marked and classified for Default forwarding (or dropped, as a last resort). A traffic policing function **SHOULD** allow approximately 100 ms of burst tolerance (e.g., a token bucket depth equal to 100 ms multiplied by the policer rate). A traffic-shaping function **SHOULD** allow approximately 10 ms of burst tolerance and no more than 50 ms of buffering. The burst tolerance values recommended here are intended to reduce the degradation that could be introduced to delay- and loss-sensitive traffic marked NQB without significantly degrading Default traffic and that could be adjusted based on local network policy. Increasing the burst tolerance would further reduce the potential for degradation (increased loss or increased delay) of traffic marked NQB but would come at the cost of an increased risk of degradation (increased loss or increased delay) of Default traffic.

The recommendation to limit NQB traffic to 5% is based on an assumption that internal links in the downstream domain could have data rates as low as one tenth of the interconnect rate; in which case, if the entire aggregate of NQB traffic traversed a single instance of such a

link, the aggregate would consume no more than 50% of that link's capacity. The limit for NQB traffic **SHOULD** be adjusted based on any knowledge of the local network environment that is available.

6.5. The NQB DSCP and Tunnels

[RFC2983] discusses tunnel models that support Diffserv. It describes a "uniform model" in which the inner DSCP is copied to the outer header at encapsulation and the outer DSCP is copied to the inner header at decapsulation. It also describes a "pipe model" in which the outer DSCP is not copied to the inner header at decapsulation. Both models can be used in conjunction with the NQB PHB. In the case of the pipe model, any DSCP manipulation (re-marking) of the outer header by intermediate nodes would be discarded at tunnel egress. In some cases, this could improve the possibility of achieving NQB treatment in subsequent nodes; in other cases, it could degrade that possibility (e.g., if the re-marking was designed specifically to preserve NQB treatment in downstream domains).

As is discussed in [RFC2983], tunnel protocols that are sensitive to reordering (such as IPsec [RFC4301] or Layer 2 Tunneling Protocol (L2TP) [RFC2661]) can result in undesirable interactions if multiple DSCP PHBs are signaled for traffic within a tunnel instance. This is true for tunnels containing a mix of QB and NQB traffic. Additionally, since networks supporting the NQB PHB could implement a traffic protection mechanism (see Section 5.2) and/or responses to NQB buffer overrun that result in out-of-order delivery for traffic marked with the NQB DSCP, even tunnels solely containing NQB traffic could have issues if they are sensitive to reordering and the outer header retains the NQB DSCP. As a result, the use of a reordering-sensitive tunnel protocol to carry NQB traffic, or a mix of QB and NQB traffic, necessitates that the outer tunnel header be re-marked with a non-NQB DSCP (e.g., Default); in this case, the "pipe" model is preferable because it preserves the marking differentiation at tunnel decapsulation.

6.6. Guidance for Lower-Rate Links

The NQB sender requirements in Section 4 place responsibility in the hands of the application developer to determine the likelihood that the application's sending behavior could result in a queue forming along the path. These requirements rely on application developers having a reasonable sense for the network context in which their application is to be deployed. Even so, there will undoubtedly be networks that contain links having a data rate that is below the lower end of what is considered "typical"; some of these links could even be below the instantaneous sending rate of some NQB-marked applications.

To limit the consequences of this scenario, operators of networks with lower rate links **SHOULD** consider utilizing a traffic protection function on those links that is more tolerant of burstiness (i.e., a temporary queue). This will have the effect of allowing a larger set of NQB-marked microflows to remain in the NQB queue; however, it will come at the expense of a greater potential for delay variation. In implementations that support [RFC9957], the burst tolerance can be configured via the CRITICALqLSCORE_us input parameter.

Alternatively, operators of networks with lower rate links **MAY** choose to disable NQB support (and thus aggregate traffic marked with the NQB DSCP with Default traffic) on these lower rate links. For links that have a data rate that is less than 10% of "typical" path rates, it is **RECOMMENDED** that the NQB PHB be disabled and that traffic marked with the NQB DSCP is therefore carried using the Default PHB (without being re-marked to the Default DSCP (0)).

7. Mapping NQB to Standards of Other SDOs

This section provides recommendations for the support of the NQB PHB in certain use cases. This section is not exhaustive.

7.1. DOCSIS Access Networks

Residential cable broadband Internet services are commonly configured with a single bottleneck link (the access network link) upon which the service definition is applied. The service definition, typically an upstream/downstream data rate tuple, is implemented as a configured pair of rate shapers that are applied to the user's traffic. In such networks, the QoS that each application receives, and as a result, the QoE that it generates for the user is influenced by the characteristics of the access network link.

To support the NQB PHB, cable broadband services would need to be configured to provide a separate queue for traffic marked with the NQB DSCP. The NQB queue would need to be configured to share the service's rate-shaped capacity with the queue for QB traffic. Further discussion about support of the NQB PHB in DOCSIS networks can be found in [\[LOW_LATENCY_DOCSIS\]](#).

7.2. Mobile Networks

Historically, 3GPP mobile networks have utilized "bearers" to encapsulate each user's user plane traffic through the radio access and core networks. Bearers are also associated with a QoS that determines how packets are prioritized at queues and radio schedulers. In LTE networks, these are Evolved Packet System (EPS) bearers, part of the EPS, which comprises the core and access network architecture. Typically, an LTE operator provides a dedicated EPS bearer for IP Multimedia Subsystem (IMS) Voice over LTE (VoLTE) traffic to meet regulatory obligations for call completion rates and a best-effort default EPS bearer for Internet traffic. This default EPS bearer is typically non-Guaranteed Bit Rate (non-GBR) and provides no guarantees; its buffering characteristics generally are not compatible with low-latency traffic. In 5G networks, similar functionality is provided using QoS flows within a PDU Session in the core network, which are mapped to Data Radio Bearers (DRBs) on the radio network. 5G systems offer more flexibility in QoS handling, allowing traffic to be treated according to type (e.g., loss-tolerant, low-latency); hence, they support more suitable treatment of Internet real-time microflows.

To support the NQB PHB, an LTE network could be configured to provide the User Equipment (UE, the subscriber's device) with a dedicated low-latency, non-GBR EPS bearer, in addition to the default EPS bearer. For example, this dedicated EPS bearer could use QCI 7 (QoS Class Identifier

7), which is typically used for low-latency, non-GBR services. Alternatively, in a 5G system, a Data Radio Bearer (DRB) with 5QI 7 (5G QoS Identifier 7, also used for low-latency traffic), could be provisioned (see Table 5.7.4-1: Standardized 5QI to QoS characteristics mapping in [SA-5G]).

A packet carrying the NQB DSCP could then be routed through this dedicated low-latency path, while packets without the NQB marking would be routed through the default bearer.

7.3. Wi-Fi Networks

Wi-Fi networking equipment compliant with 802.11e/n/ac/ax [IEEE802-11] generally supports either four or eight transmit queues and four sets of associated EDCA parameters (corresponding to the four Wi-Fi Multimedia (WMM) Access Categories) that are used to enable differentiated medium access characteristics. The four WMM Access Categories, referred to as Voice Access Category (AC_VO), Video Access Category (AC_VI), Best Effort Access Category (AC_BE), and Background Access Category (AC_BK), provide four levels of prioritized access to the wireless medium. As discussed in [RFC8325], it has been a common practice for Wi-Fi implementations to use a default DSCP to User Priority (UP) mapping that utilizes the most significant three bits of the DSCP field to select "User Priority", which is then mapped to the four WMM Access Categories. [RFC8325] also provides an alternative mapping that more closely aligns with the DSCP recommendations provided by the IETF. In the case of some managed Wi-Fi equipment, this mapping can be controlled by the network operator, e.g., via TR-369 [TR-369].

In addition to the requirements provided in other sections of this document, to support the NQB PHB, Wi-Fi equipment (including equipment compliant with [RFC8325]) **SHOULD** map the DSCP value 45 (decimal) into a separate queue in the same Access Category as the queue that carries Default traffic (i.e., the Best Effort Access Category). It is **RECOMMENDED** that Wi-Fi equipment provide a separate queue in UP 0 and map the DSCP value 45 (decimal) to that queue. If a separate queue in UP 0 cannot be provided (due to hardware limitations, etc.), a Wi-Fi device **MAY** map the DSCP value 45 (decimal) to UP 3.

7.3.1. Interoperability with Existing Wi-Fi Networks

While some existing Wi-Fi equipment might be capable (in some cases via firmware update) of supporting the NQB PHB requirements, many currently deployed devices cannot be configured in this way. As a result, the remainder of this section discusses interoperability with these existing Wi-Fi networks, as opposed to PHB compliance.

Since this equipment is widely deployed, and the Wi-Fi link can become a bottleneck link, the performance of traffic marked with the NQB DSCP across such links could have a significant impact on the viability and adoption of the NQB DSCP and PHB. Depending on the DSCP used to mark NQB traffic, existing Wi-Fi equipment that uses the default mapping of DSCPs to Access Categories (see Section 2.3 of [RFC8325]) and the default EDCA parameters will support either (but not both) of the following characteristics:

- the NQB PHB requirement for separate queuing of NQB traffic from Default traffic (Section 5.1)
- the recommendation to treat NQB traffic with forwarding preference equal to that used for Default traffic (Section 5.1)

The DSCP value 45 (decimal) is recommended for NQB (see [Section 4](#)). This maps NQB to UP 5 using the default mapping, which is in the Video Access Category. While this choice of DSCP enables these Wi-Fi systems to support the NQB PHB requirement for separate queuing, existing Wi-Fi devices generally utilize EDCA parameters that result in statistical prioritization of the Video Access Category above the Best Effort Access Category. In addition, this equipment does not support the remaining NQB PHB recommendations in [Section 5](#). The rationale for the choice of DSCP value 45 (decimal) as well as its ramifications and remedies for its limitations are discussed further below.

The choice of separated queuing rather than equal forwarding preference in existing Wi-Fi networks was motivated by the following:

- Separate queuing is necessary in order to provide a benefit for traffic marked with the NQB DSCP.
- The arrangement of queues in Wi-Fi equipment is typically fixed, whereas the relative priority of the Access Category queues is configurable. Most Wi-Fi equipment has hardware support (albeit generally not exposed for user control), which could be used to adjust the EDCA parameters in order to meet the equal forwarding preference recommendation. This is discussed further below.
- Traffic that is compliant with the NQB sender requirements in [Section 4](#) is expected to cause minimal degradation to traffic in lower priority Access Categories. In any case, it would be unlikely to cause more degradation to lower priority Access Categories than the existing recommended Video Access Category traffic types: Broadcast Video, Multimedia Streaming, Multimedia Conferencing from [[RFC8325](#)], and AudioVideo, ExcellentEffort from [[QOS_TRAFFIC_TYPE](#)].
- Several existing client applications that are compatible with the NQB sender requirements already select the Video Access Category; thus, they would not see a degradation in performance by transitioning to the NQB DSCP, regardless of whether the network supported the PHB.
- Application instances on Wi-Fi client devices are already free to choose any Access Category that they wish, regardless of their sending behavior, without any policing of usage. So, the choice of using DSCP value 45 (decimal) for NQB creates no new avenues for non-NQB-compliant client applications to exploit the prioritization function in Wi-Fi.
- For application traffic that originates outside of the Wi-Fi network, and, thus, is transmitted by the Access Point, the choice of DSCP value 45 (decimal) does create a potential for abuse by non-compliant applications. However, opportunities exist in the network components upstream of the Wi-Fi Access Point to police the usage of the NQB DSCP and potentially re-mark traffic that is considered non-compliant, as is recommended in [Section 6.4.1](#). Furthermore, it is reasonable to expect that ISPs currently manage the DSCPs on traffic destined to their customers' networks and will continue to do so whether or not they support NQB. This includes the practice in residential broadband networks of re-marking the DSCP field to zero on all traffic. Any change to these practices done to enable the NQB DSCP to pass through could be done alongside the implementation of the recommendations in [Section 6.4.1](#).

The choice of Video Access Category rather than the Voice Access Category was motivated by the desire to minimize the potential for degradation of Best Effort Access Category traffic. The choice of Video Access Category rather than the Background Access Category was motivated by the much greater potential of degradation to NQB traffic that would be caused by the vast majority of traffic in most Wi-Fi networks, which utilizes the Best Effort Access Category.

As stated above, the use of DSCP value 45 (decimal) for NQB is not expected to create incentives for abuse by non-compliant applications in the Wi-Fi uplink direction. The fact that the NQB DSCP brings with it the potential for degradation of non-compliant applications (traffic protection and/or a shallow queue resulting in reordering and/or packet loss at some hop along the path) plus the existence of multiple other DSCP values that don't carry the risk of degradation and that could be readily used to obtain prioritization (AC_VI or even AC_VO) leads to the conclusion that NQB non-compliant applications that are seeking prioritization in the Wi-Fi uplink would be better off selecting one of those other DSCPs. This conclusion is not expected to be disturbed by network support for NQB increasing the likelihood of DSCP value 45 (decimal) traffic traversing network boundaries without change to the DSCP, as that likelihood of increased network boundary traversal is balanced by a likelihood of NQB traffic encountering the traffic-limiting aspects of NQB support, traffic protection, and shallow buffers, which limit the potential for abuse.

In the case of traffic originating outside of the Wi-Fi network, the prioritization of traffic marked with the NQB DSCP via the Video Access Category (if left unchanged) could potentially erode the principle of alignment of incentives discussed in [Section 5](#). In order to preserve the incentives principle for NQB, Wi-Fi systems **MAY** be configured such that the EDCA parameters for the Video Access Category match those of the Best Effort Access Category, which will mean AC_VI is at the same priority level as AC_BE. These changes might not be possible on all Access Points; in any case, the requirements and recommendations in [Section 6.4.1](#) would apply in this situation.

Systems that utilize [\[RFC8325\]](#) but cannot provide a separate AC_BE queue for NQB traffic **SHOULD** map the DSCP value 45 (decimal) (or the locally determined alternative) to UP 5 in the Video Access Category as well (see [Section 7.3.2](#)).

7.3.2. The Updates to RFC 8325

[Section 4.2.9](#) of [\[RFC8325\]](#) describes the recommendation for the handling of Standard service class traffic that carries the Default DSCP. This update to [\[RFC8325\]](#) changes the title of [Section 4.2.9](#) of [\[RFC8325\]](#) from "Standard" to "Standard and Non-Queue-Building". This update additionally adds a paragraph at the end of [Section 4.2.9](#) of [\[RFC8325\]](#) as follows:

RFC 9956 defines a shallow-buffered, best-effort service for traffic described as Non-Queue-Building and recommends the following treatment in Wi-Fi equipment. It is **RECOMMENDED** that Wi-Fi equipment map Non-Queue-Building traffic into a separate queue in the same Access Category as the queue that carries Default traffic (i.e., the Best Effort Access Category). It is **RECOMMENDED** that Wi-Fi equipment provide a separate queue in UP 0 and map Non-Queue-Building traffic to that queue. If a separate queue in UP 0 cannot be provided (due to hardware limitations, etc.), a Wi-Fi device **MAY** map

Non-Queue-Building traffic to UP 3. If neither of these options provides a separate queue from Default traffic, it is **RECOMMENDED** that Wi-Fi equipment map Non-Queue-Building traffic to UP 5 (which corresponds to the default mapping described in Section 2.3). RFC 9956 provides additional recommendations and requirements for support of the NQB PHB that aren't available in the QoS model described in Section 6 but nonetheless could be supported in implementations.

In another update to [RFC8325] captured below, a new row for "Non-Queue-Building" traffic is inserted between the existing "Low-Latency Data" and "OAM" rows in Figure 1 of [RFC8325] as follows:

IETF Diffserv Service Class	PHB	Reference RFC	IEEE 802.11	
...	...	...	User Priority	Access Category
Low-Latency Data	AF21 AF22 AF23	RFC 2597	3	AC_BE (Best Effort)
Non- Queue- Building	NQB	RFC 9956	0, 3 5 See Section 4.2.9	AC_BE (Best Effort) OR AC_VI (Video)
OAM	CS2	RFC 2474	0	AC_BE (Best Effort)

A third update adds the following sentence to the end of the first paragraph in Section 5.3 of [RFC8325]:

An exception to this is the NQB DSCP value 45 (decimal), which encodes for best-effort service.

8. IANA Considerations

IANA has assigned the Differentiated Services Field Codepoint (DSCP) 45 from the "Differentiated Services Field Codepoints (DSCP)" registry (<https://www.iana.org/assignments/dscp-registry/>) ("DSCP Pool 3 Codepoints", Codepoint Space xxxx01, Standards Action ([RFC8126]) for NQB behavior.

IANA has updated this registry as follows:

Name: NQB

Value (Binary): 101101

Value (Decimal): 45

Reference: RFC 9956

9. Security Considerations

The security considerations for the NQB PHB relate to the potential to impact the capacity available or delay experienced by other flows that share a bottleneck on the path with traffic that is marked with the recommended NQB DSCP.

Full support for the NQB PHB in bottleneck links limits the incentives for a QB application to mis-mark its packets as NQB, particularly for implementations that support traffic protection. If a QB microflow were to mis-mark its packets as NQB, it would be unlikely to receive a benefit by doing so, and it would usually experience a degradation, in contrast to mis-marking its packets for a higher-priority PHB, e.g., the EF PHB [RFC3246]. The nature of the degradation would depend on the specifics of the PHB implementation, including response to NQB buffer overflow (and on the presence or absence of a traffic protection function) but could include excessive packet loss, excessive delay variation, and/or excessive out-of-order delivery. If an NQB microflow were to fail to mark its packets as NQB, it could suffer the delay and loss typical of sharing a queue with capacity-seeking traffic.

To preserve low-delay performance for NQB traffic, networks that support the NQB PHB will need to ensure that mechanisms are in place to prevent malicious traffic marked with the NQB DSCP from causing excessive queue delay. [Section 5.2](#) recommends the implementation of a traffic protection mechanism to achieve this goal. The recommendations on traffic protection mechanisms in this document presume that some type of "flow" state be maintained in order to differentiate between microflows that are causing queuing delay and those that aren't. Since this flow state is likely finite, this opens up the possibility of flow-state exhaustion attacks. While this document requires that traffic protection mechanisms be designed with this possibility in mind, the outcomes of flow-state exhaustion would depend on the implementation.

If traffic protection is not implemented or is not able to prevent queue formation in the NQB shallow buffer, the limited size of that buffer will cause a growing queue to overrun that buffer, resulting in negative effects (e.g., reforwarding as Default, discarding) that potentially impact multiple NQB-marked microflows, independent of whether each affected microflow contributed to queue formation. As discussed elsewhere in this document, those negative effects serve to discourage misuse and abuse of NQB by QB traffic, but the negative side effects on NQB traffic that is using NQB (and the associated shallow buffer) as intended motivates limiting the effects of shallow buffer overrun via per-user provisioning limits that prevent queue overrun effects from affecting other users (see [Section 5.1](#)).

Notwithstanding the above, the choice of DSCP for NQB does allow existing Wi-Fi networks to readily (and by default) support some of the PHB requirements, but without a traffic protection function, and (when left in the default state) by giving NQB traffic higher priority than QB traffic.

This is not considered to be a compliant implementation of the PHB. These existing Wi-Fi networks currently provide priority to half of the DSCP space, whether or not the DSCP value 45 (decimal) is assigned to the NQB DSCP. While the DSCP value 45 (decimal) could also be abused to gain priority on such links, the potential presence of traffic protection functions in other hops along the path (which likely act on the DSCP value 45 (decimal) alone) would make it less attractive for such abuse than any of the other 31 DSCP values that are given priority.

This document discusses the potential use of the NQB DSCP and NQB PHB in network technologies that are standardized in other SDOs. Any security considerations that relate to deployment and operation of NQB solely in specific network technologies are not discussed here.

NQB uses the DS field. The design of Diffserv does not include integrity protection for the DSCP; thus, it is possible for the DSCP to be changed by an on-path attacker. The NQB PHB and associated DSCP don't change this. While re-marking DSCPs is permitted for various reasons (some are discussed in this document, others can be found in [RFC2474] and [RFC2475]), if done maliciously, this might negatively affect the QoS of the tampered microflow. Nonetheless, an on-path attacker can also alter other mutable fields in the IP header (e.g., the TTL), which can wreak much more havoc than just altering QoS treatment.

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Appendix A. DSCP Re-marking Policies

Some network operators typically bleach (zero out) the DSCP field on ingress into their network (see [RFC9435], [Custura], and [Barik]) and, in some cases, apply their own DSCP for internal use. Bleaching the DSCP value 45 (decimal) is not expected to cause harm to Default traffic, but it will severely limit the ability to provide NQB treatment. Reports on existing deployments of DSCP manipulation (see [Custura] and [Barik]) categorize the re-marking behaviors into the following policies: bleach all traffic (set DSCP to zero); set the top three bits (the former Precedence bits) on all traffic to 0b000, 0b001, or 0b010; set the low three bits on all traffic to 0b000; or re-mark all traffic to a particular (non-zero) DSCP value.

Regarding the DSCP value 45 (decimal), there were no observations of DSCP manipulation reported in which traffic was marked with the DSCP value 45 (decimal) by any of these policies. Thus, it appears that these re-marking policies would be unlikely to result in QB traffic being marked as NQB. In terms of the fate of traffic marked with the DSCP value 45 (decimal) that is subjected to one of these policies, it would be indistinguishable from some subset (possibly all) of other traffic. In the policies where all traffic is re-marked using the same (zero or non-zero) DSCP value, the ability for a subsequent network hop to differentiate NQB traffic via DSCP would clearly be lost entirely.

In the policies where the top three bits are overwritten (see Section 4.2 of [RFC9435]), the DSCP value 45 (decimal) would receive the same marking as would the currently unassigned Pool 3 DSCP values (5, 13, 21, 29, 37, 53, and 61), with all of these DSCP values getting re-marked to DSCP value = 5, 13, or 21 (depending on the overwrite value used). Since none of the DSCP values in the preceding lists are currently assigned by IANA, and they all are reserved for Standards Action, it is believed that they are not widely used currently; however, this could vary based on local-usage and could change in the future. If networks in which this sort of re-marking occurs or networks downstream classify the resulting DSCP (i.e., 5, 13, or 21) to the NQB PHB or re-mark such traffic with DSCP value 45 (decimal), they risk giving NQB treatment to other traffic that was not originally marked as NQB. In addition, as described in Section 6 of [RFC9435] future assignments of these 0bxxx101 DSCPs would need to be made with consideration of the potential that they all are treated as NQB in some networks.

For the policy in which the low three bits are set to 0b000, the DSCP value 45 (decimal) would be re-marked to CS5 and would be indistinguishable from CS5, VA, and EF (and the currently unassigned DSCP values 41, 42, and 43). Traffic marked using the existing standardized DSCPs in this list are likely to share the same general properties as NQB traffic (non-capacity-seeking and very low data rate, relatively low data rate, and consistent data rate). Similarly, any future recommended usage for DSCP values 41, 42, 43 would likely be somewhat compatible with NQB treatment, assuming that IP Precedence compatibility (see [Section 1.5.4](#) of [\[RFC4594\]](#)) is maintained in the future. Here there might be an opportunity for a node to provide the NQB PHB or the CS5 PHB to CS5-marked traffic and retain some of the benefits of NQB marking. This could be another motivation to classify CS5-marked traffic into the NQB queue (as discussed in [Section 6.3](#)).

Appendix B. Comparison with Expedited Forwarding

The EF definition [\[RFC3246\]](#) provides the following text to describe the EF PHB forwarding behavior:

This specification defines a PHB in which EF packets are guaranteed to receive service at or above a configured rate

and

the rate at which EF traffic is served at a given output interface should be at least the configured rate R , over a suitably defined interval, independent of the offered load of non-EF traffic to that interface.

Notably, this description is true of any class of traffic that is configured with a guaranteed minimum rate, including the Default PHB if configured per the guidelines in [Section 1.5.1](#) of [\[RFC4594\]](#). [\[RFC3246\]](#) goes on to formalize the definition of EF by requiring that an EF node be characterizable in terms of the fidelity with which it is able to provide a guaranteed rate.

While the NQB PHB is not required to be configured with a guaranteed minimum rate, [\[RFC2474\]](#) and [\[RFC4594\]](#) recommend assigning some minimum resources for the Default PHB, in particular, some dedicated capacity. If such a guaranteed minimum rate is configured for the Default PHB, it is recommended ([Section 5](#)) that NQB traffic share and be given equal access to that rate. In such cases, the NQB PHB could effectively receive a rate guarantee of (for example) 50% of the rate guaranteed to the combined NQB/Default PHBs; therefore, it technically complies with the PHB forwarding behavior defined for EF.

However, EF is intended to be a managed service and requires that traffic be policed such that the arriving rate of traffic into the EF PHB doesn't exceed the guaranteed forwarding rate configured for the PHB. This ensures that low delay and low-delay variation are provided. NQB is intended as a best effort service; hence, the aggregate of traffic arriving to the NQB PHB queue

could exceed the forwarding rate available to the PHB. [Section 5.2](#) discusses the recommended mechanism for handling excess traffic in NQB. While EF relies on rate policing and dropping of excess traffic at the domain border, this is only one option for NQB. NQB primarily recommends traffic protection located at each potential bottleneck, where actual queuing can be detected and where excess traffic can be reclassified into the Default PHB rather than dropping it. Local traffic protection is more feasible for NQB, given the focus is on access networks, where one node is typically designed to be the known bottleneck where traffic control functions all reside. In contrast, EF is presumed to follow the Diffserv architecture [[RFC2475](#)] for core networks, where traffic conditioning is delegated to border nodes in order to simplify high-capacity interior nodes. Further, NQB recommends a microflow-based mechanism to limit the performance impact of excess traffic to those microflows causing potential congestion of the NQB queue, whereas EF ignores microflow properties. Note that, under congestion, low loss for NQB-conformant flows is only ensured if such a mechanism is operational. Note also that this mechanism for NQB operates at the available forwarding rate for the PHB (which could vary based on other traffic load) as opposed to a configured guaranteed rate, as in EF.

The lack of a requirement of a guaranteed minimum rate, and the lack of a requirement to police incoming traffic to such a rate, makes the NQB PHB suitable for implementation in networks where link capacity is not or cannot be guaranteed.

There are additional distinctions between EF and NQB arising from the intended usage as described in [[RFC4594](#)] and the actual usage in practice in the Internet. In [Section 1.5.3](#) of [[RFC4594](#)], EF is described as generally being used to carry voice or data that requires "wire-like" behavior through the network. The NQB PHB similarly is useful to carry application traffic requiring wire-like performance, characterized by low delay and low-delay variation, but places a pre-condition that each microflow be relatively low data rate and sent in a smooth (non-bursty) manner. In actual practice, EF traffic is oftentimes prioritized over Default traffic. This contrasts with NQB traffic, which is to be treated with the same forwarding precedence as Default (and sometimes aggregated with Default).

Appendix C. Impact on Higher Layer Protocols

The NQB PHB itself has no impact on higher layer protocols because it only isolates NQB traffic from QB traffic. However, traffic protection of the PHB can have unintended side effects on higher layer protocols. Traffic protection introduces the possibility that microflows classified into the NQB queue could experience out-of-order delivery or packet loss if their behavior is not consistent with the NQB sender requirements. Out-of-order delivery could be particularly likely if the traffic protection algorithm makes decisions on a packet-by-packet basis. In this scenario, a microflow that is (mis-)marked as NQB and that causes a queue to form in this bottleneck link could see some of its packets forwarded by the NQB queue and some of them either discarded or redirected to the QB queue. In the case of redirection, depending on the queuing delay and scheduling within the network element, this could result in packets being delivered out of order. As a result, the use of the NQB DSCP by a higher layer protocol carries some risk that an increased amount of out-of-order delivery or packet loss will be experienced. This characteristic provides one disincentive for incorrectly setting the NQB DSCP on traffic that doesn't comply with the NQB sender requirements.

Appendix D. Alternative Diffserv Codepoints

In networks where the DSCP value 45 (decimal) is already in use for another (e.g., a local-use) purpose or where specialized PHBs are available that can meet specific application requirements (e.g., a guaranteed-delay path for voice traffic), use of another DSCP value could be preferred.

In end systems where the choice of using DSCP value 45 (decimal) is not available to the application, the CS5 DSCP (40 decimal) could be used as a fallback. See [Section 6.3](#) for rationale as to why this choice could be fruitful.

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Authors' Addresses

Greg White

CableLabs

Email: g.white@cablelabs.com**Thomas Fossati**

Linaro

Email: thomas.fossati@linaro.org**Rüdiger Geib**

Deutsche Telekom

Email: Ruediger.Geib@telekom.de